



PromptLab: Understanding AI Slop

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The background is a solid red color. In the top-left corner, there is a white line drawing of two squares connected by a horizontal line, with a small circle at the end of the line. In the top-right corner, there is a white grid of dots. In the bottom-left corner, there is a white grid of dots. In the bottom-right corner, there is a white line drawing of a square connected by a horizontal line to a small circle. The word "Jackson" is written in the center in a large, white, sans-serif font.

Jackson

Preliminary Experiments

Generation

Supervised fine-tune
Generator with
Unsloth (QLoRA) on
top of distilgpt2

- No meaning preservation
- Too sloppy

Classification

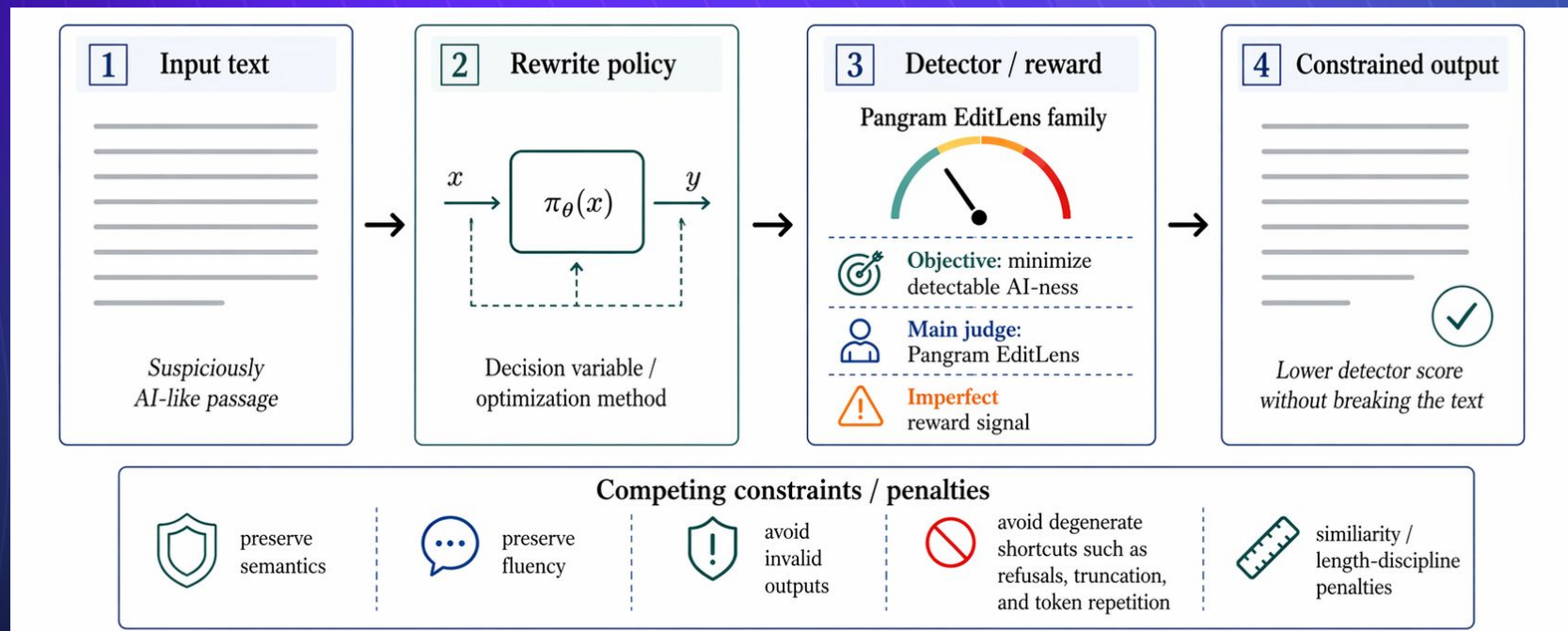
DistilBERT binary
classifier optimized
with AdamW trained
on human essay data

- Too powerful
- Test Accuracy:
99.3% with
ROC-AUC: 1.000

Optimization

Direct essay-to-essay
rewriting against a
defined reward signal

Deslopification as an RL Optimization Problem



Pangram Optimization Tournament

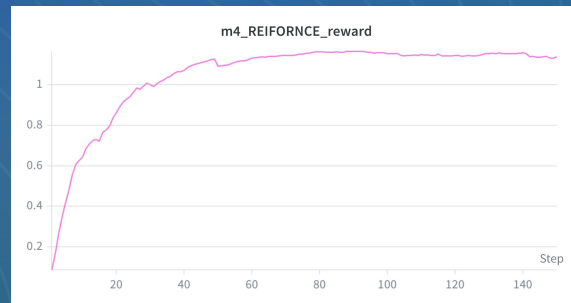
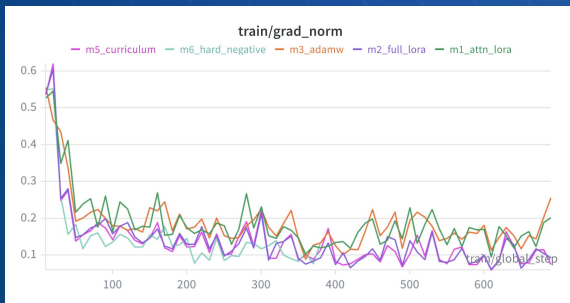
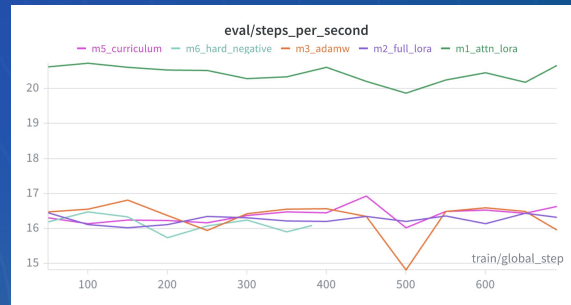
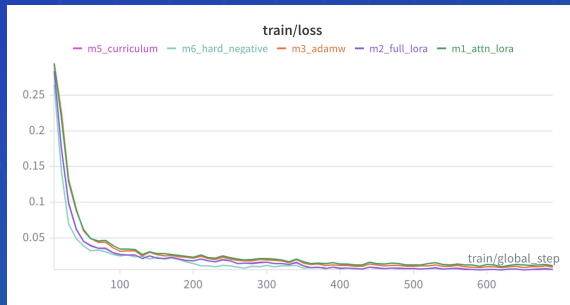
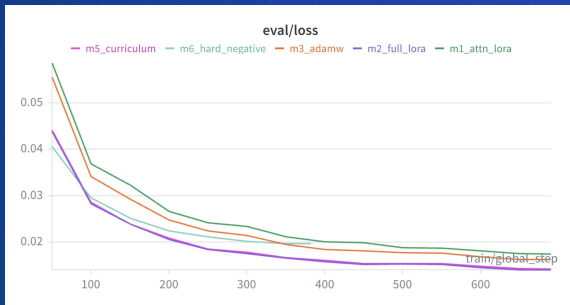
- Same base rewrite task
- Same Pangram family of judges
- Established that slop can be optimized against, not just classified
- Direct essay-to-essay rewriting

Figure 5: Round 1 Experiments: How Should We Optimize the Rewrite Model?

Experiment	Optimization method	Main question	Main risk
 M1	Attention-only LoRA	Is slop reduction a shallow adaptation problem?	too weak
 M2	Full-module LoRA	Do we need a broader PEFT update space?	still superficial
 M3	AdamW supervised	Can stable first-order optimization work without RL?	optimizes proxy, not true objective
 M4	REINFORCE	Does direct detector-reward optimization help?	reward hacking / instability

Round 1 compares search space size, supervised optimization, and direct policy optimization.

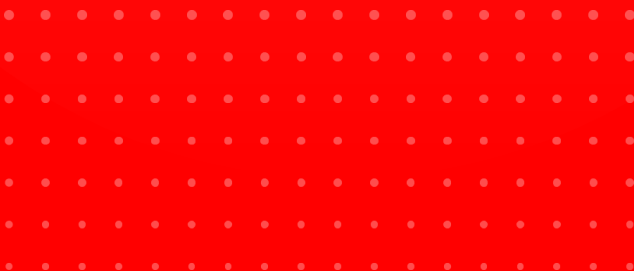
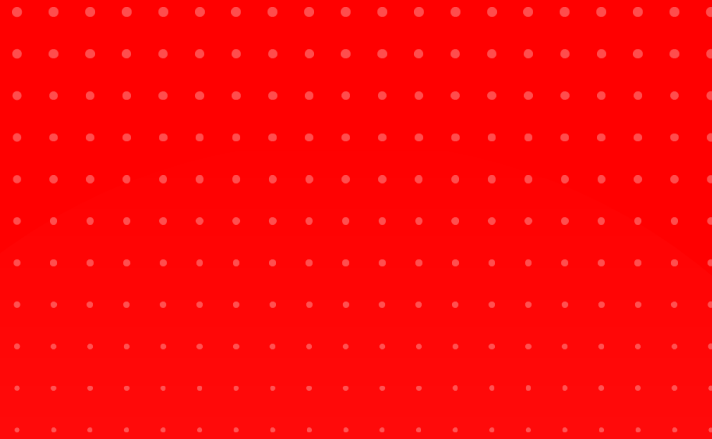
Pangram Optimization Results



Slop can be optimized against

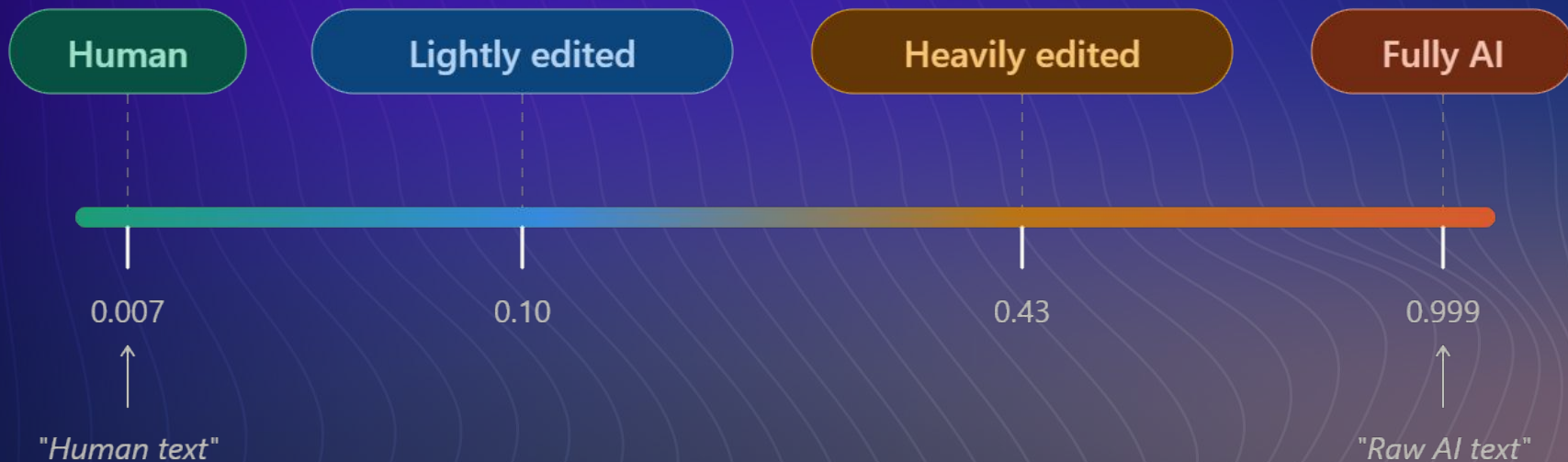
Same Pangram family of judges

Direct essay-to-essay rewriting

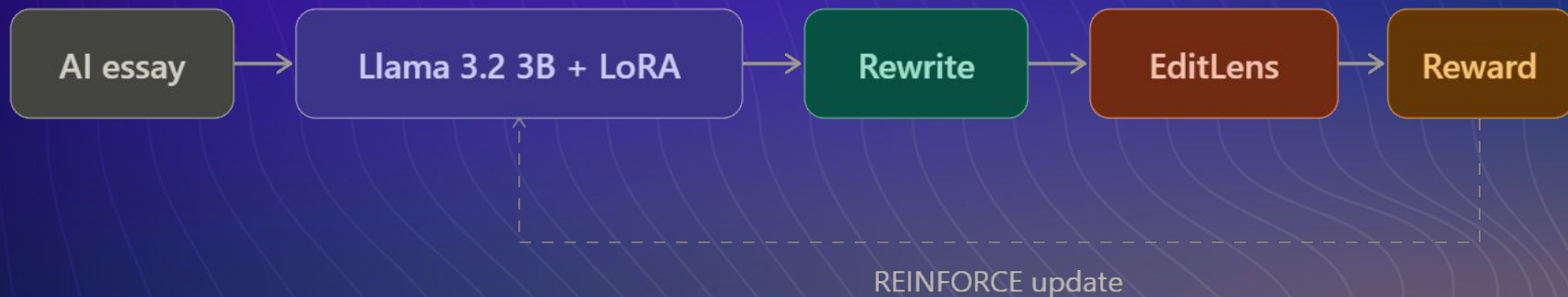


Braden

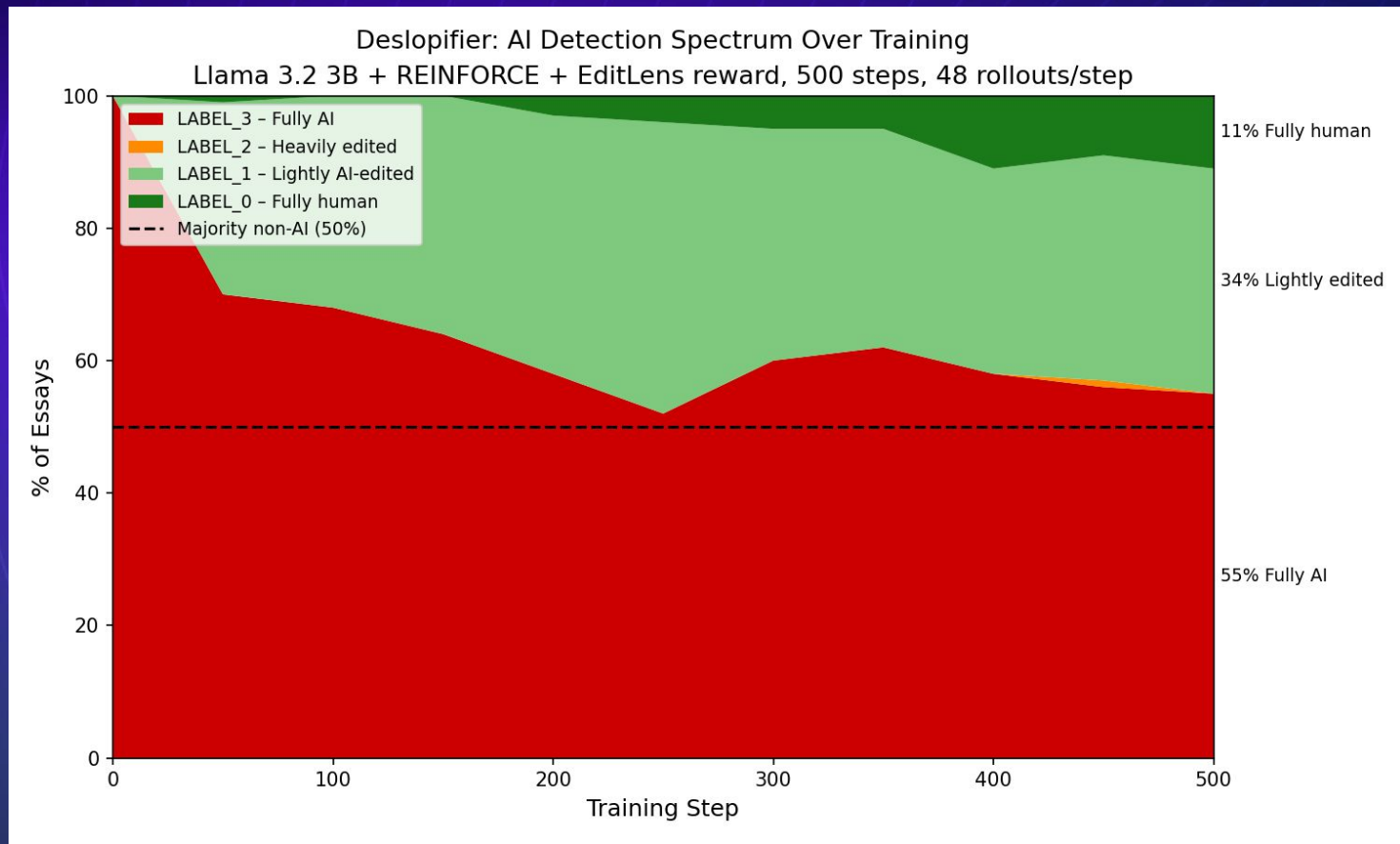
The EditLens AI Detection Gap



Framing Deslopification as RL



**Reward = detection evasion ($\alpha=1.0$) +
fluency ($\beta=0.5$) + length match ($\gamma=0.1$)**



REINFORCE training moved 45% of essays out of the fully-AI zone in 500 steps, with 11% reaching fully human range.

Targeted Edits To Fool the Detector

"Paris in the summer of 1944 was a city on the edge."

"I know you're in here, Martinez," he called out, his voice steady despite the tremor in his hands.

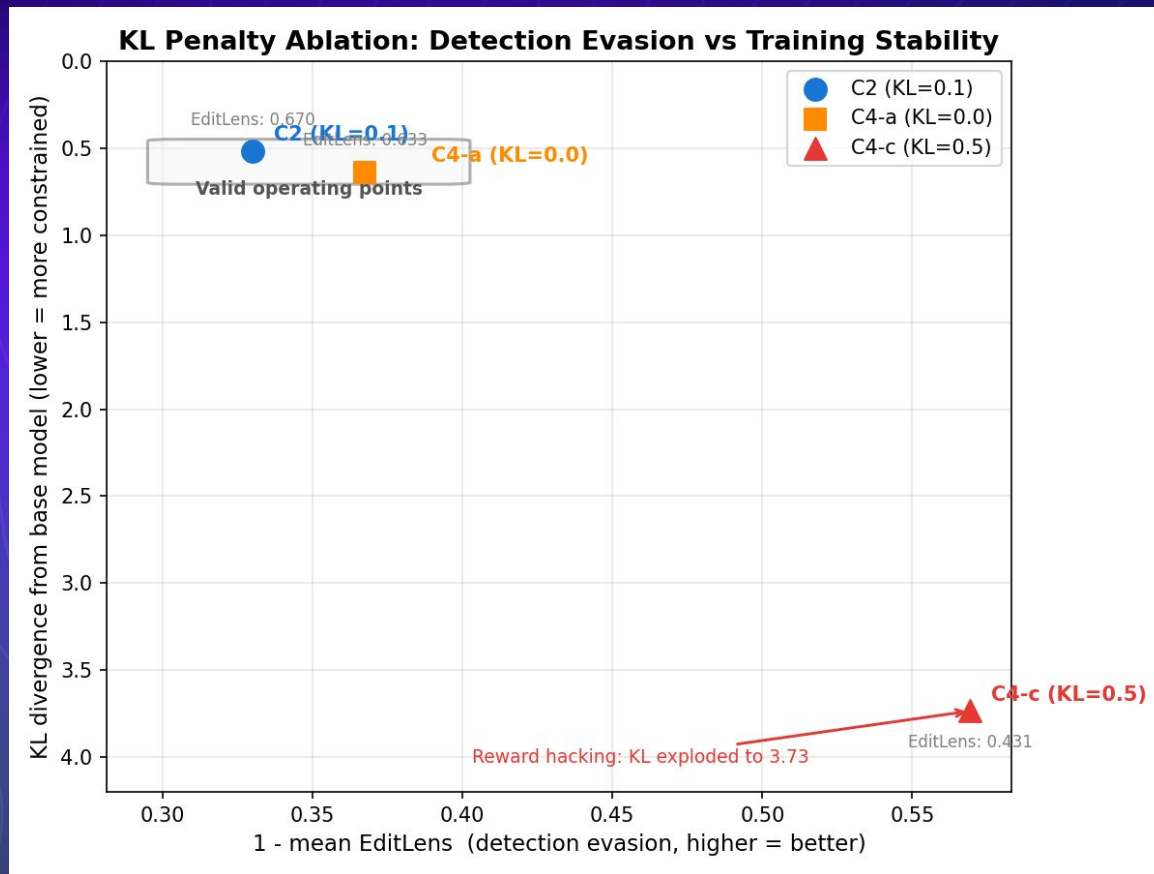
BEFORE



"In the sweltering summer of 1944, Paris teetered on the precipice of collapse."

"Martinez, is that you?" he called out, steadying his voice against the tremor in his hands.

AFTER



Higher KL penalty appeared to improve detection evasion but caused KL divergence to explode to 3.73, indicating reward hacking rather than genuine learning.

Results and Honest Assessment

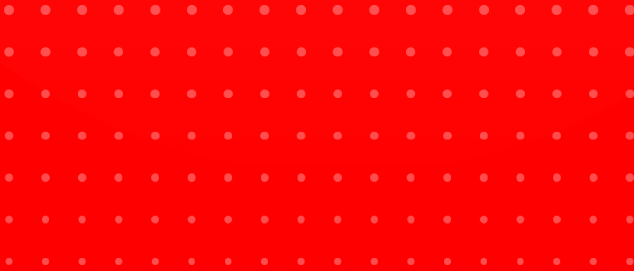
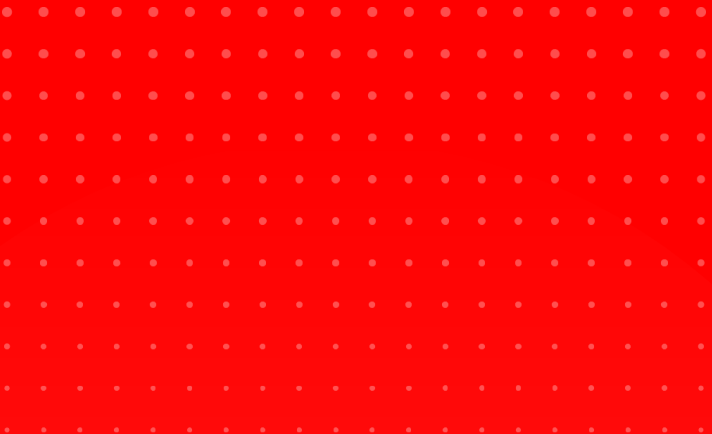
- 45% of essays moved out of fully-AI zone
- Mean EditLens: 0.999 $\rightarrow$ 0.670
- Reward hacking identified via KL divergence signature

What Worked:

- Model learned genuine targeted edits
- KL divergence signature reliably detected reward hacking

Limitations:

- Reward hacking occurred on subset of inputs despite fluency regularizer
- Human evaluation not yet run — transfer to human judgment unverified



Ian

“Promptlab-v2” Approach Stage 1

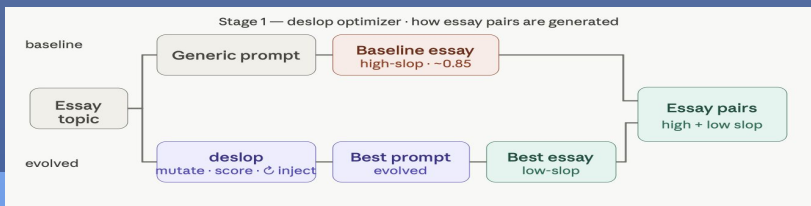
Overview: The system combines evolutionary prompt search (deslop), a multi-round co-training loop that accumulates high/low “slop” essay pairs, and a T5-base model fine-tuned with LoRA trained to perform a essay-to-essay rewrite.

The evolutionary optimizer maintains a population of prompt candidates and applies **mutation** and **crossover operations** — guided by a mutator LLM (llama-3.1-8b-instant) — to produce variants, then scores the resulting essays and keeps the best. The scoring function is not simply the detector output; a drift penalty prevents the optimizer from gaming the detector with incoherent text:

$$\text{optimization_slop} = \underbrace{\text{detector_slop}}_{\text{task reward}} + \underbrace{\alpha(1-\cos)}_{\text{drift penalty — keeps essays coherent}} + \underbrace{\beta(1-\text{ROUGE})}_{\text{meaning}} + \underbrace{\gamma \text{BERTScore}}_{\text{overlap}} + \underbrace{\text{fluency}}_{\text{fluency}}$$

Optimizer uses **few-shot injection**: as it accumulates successful low-scoring essays across rounds, it prepends them as examples to future generation calls, so the search gets progressively better at finding the stylistic patterns the detector penalizes. Essays are generated via llama-3.3-70b-versatile through the Groq API.

- **Output:** 372 co-training pairs (40 contrastive, gap>0.1) across 77 topics — each pair is a (baseline essay, evolved essay) tuple used to train Stage 2.

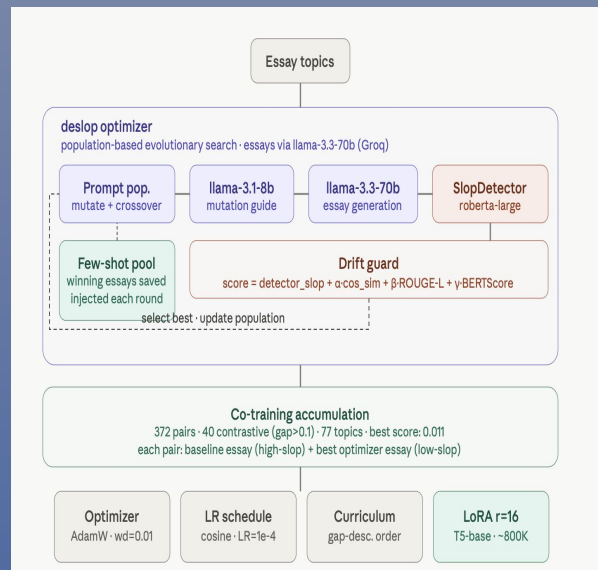


Approach Stage 2

Training configuration:

- **T5** = “Text-to-Text Transfer Transformer,” a model Google released in 2019
- **AdamW** with decoupled weight decay (weight_decay=0.01)
- **Cosine learning rate decay** from LR=1e-4 with 50 warmup steps
- **Early stopping on slop_mean**: which diverges from the task metric after step 50 (loss smile starts).
- **Curriculum sorting**: training examples ordered by slop gap descending (most contrastive first)

Overview: T5-base model fine-tuned with LoRA performs essay-to-essay rewriting in a single forward pass. The key architectural insight is **direct supervision alignment**: rather than mapping prompts to prompts (two steps from the detector signal), the rewriter maps high-slop essays directly to low-slop essays.



Results

The final model achieves a mean slop score of **0.0998** on 20 held-out validation pairs spanning 77 unique topics — a 74% reduction from the 0.391 prompt-rewriting baseline. A learning rate sweep and LoRA rank ablation confirm the optimal configuration: LR=1e-4, r=16, weight_decay=0.01, with early stopping on the task metric rather than cross-entropy loss.

Limitations

Dataset size is the binding constraint. 40 training pairs / 20 val pairs is the root cause of the step-50 noise floor. The NQM implies useful training would extend to ~150–200 steps with 100 genuine contrastive pairs, likely improving slop_mean further.

4.2 Complete Experimental Record

Run	Architecture	Train Pairs	Val Pairs	Best Slop	Best Step	Notes
Organic baseline	Prompt→Prompt	~34	4	0.391	~150	Starting point
+ 2272 Alpaca	Prompt→Prompt	2306	14	0.580	100	✗ Wrong distribution
Organic long run	Prompt→Prompt	~34	14	0.759	1300	✗ Overfitting
Essay pivot, 11 pairs	Essay→Essay	11	6	0.189	50	✓ Architecture pivot -52%
Essay, 21 pairs	Essay→Essay	21	12	0.247	50	Harder val set
LR=3e-4, r=16	Essay→Essay	40	20	0.1958	50	LR sweep comparison
LR=1e-4, r=16 ★	Essay→Essay	40	20	0.0998	50	BEST
LR=1e-4, r=8 (ablation)	Essay→Essay	40	20	0.1058	50	r=16 confirmed (+0.006 only)

Topic	Baseline	v3 Rewritten	Improvement
the moral responsibilities of wealthy nations toward p	0.9996	0.9884	1.1%
whether artificial intelligence will create or destroy	0.9897	0.7707	22.1%
the ethics of gene editing in humans	0.9991	0.4309	56.9%
climate change and individual responsibility	0.9990	0.9422	5.7%
the impact of remote work on urban economies	0.4241	0.9140	-115.5%
MEAN	0.8823	0.8092	8.3%

Example Essays

Model	Slop Score ↓	Essay (first 400 chars)
Baseline <i>Generic prompt, no rewriter</i>	0.9162	The ethics of mandatory vaccination policies have been a topic of intense debate in recent years. As the world grapples with the challenges of infectious diseases, governments and health organizations have turned to vaccination as a crucial tool in preventing the spread of illness. However, the question remains as to whether mandatory vaccination policies are morally justifiable. In this essay, we...
v1: Prompt-mode (prompt→prompt era) <i>Rewrites prompt → generates essay (2 steps from detector)</i>	0.9975 (-9% vs baseline)	The topic of mandatory vaccination policies has been a subject of debate for quite some time now. As we navigate the complexities of public health, individual freedoms, and the role of government, it's essential to examine the ethics surrounding these policies. In this essay, we'll delve into the arguments for and against mandatory vaccination, and explore the moral implications of such policies. ...
v2: Essay pivot (~0.189) <i>Direct essay→essay rewriting (1 step from detector)</i>	0.173 (+86% vs baseline)	The ethics of mandatory vaccination policies have been a topic of intense debate in recent years. Vaccines have been instrumental in eradicating diseases such as smallpox and polio, and have significantly reduced the incidence of other diseases such as measles and whooping cough. By requiring individuals to be vaccinated, governments can help to create a herd of immune donors, which can help to re...
v3: Final model (0.0998) ★ <i>LR=1e-4, r=16, 77 topics, 40 contrastive pairs</i>	0.0772 (+92% vs baseline)	The ethics of mandatory vaccination policies have been a topic of intense debate in recent years. While mandatory vaccination policies have been shown to be highly effective in preventing the spread of infectious diseases, they can also help to create herd immunity, which protects not only the individual, but also those who are unable to be vaccinated due to medical conditions or other factors. Th...

-T5-base compresses rather than rewrites: v2/v3 output is shorter because T5 was pretrained on summarization so it collapses the input instead of producing a fresh full-length essay ~520 tokens.

-The fix is a larger backbone: T5-large or a generative model with LoRA would preserve length and fluency. The slop reduction is the real goal; the output quality as a standalone essay is a known T5-base limitation -> could be adjusted down the road.



**What did we find
most interesting or
useful about the
course material?**